

CodeAlpha C Programming Internship

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Task 1: Basic Calculator Program

Aim:

To write a C program to perform basic arithmetic operations (Addition, Subtraction, Multiplication & Division).

Program:

```
#include <stdio.h>

int main() {
    double num1, num2, result;
    char operation;
    int choice;
    printf("Basic Calculator\n");
    printf("Select an operation:\n");
    printf(" 1. Addition(+)\n");
    printf(" 2. Subtraction(-)\n");
    printf(" 3. Multiplication(*)\n");
    printf(" 4. Division(/)\n");
    printf("Enter your choice (1-4):");
    if (scanf("%d", &choice) != 1) {
        printf("Invalid input. Please enter a number.\n");
        return 1;
    }
    printf("Enter first number: ");
    if (scanf("%lf", &num1) != 1) {
```

```
    printf("Invalid input. Please enter a numeric value.\n");
    return 1;
}

printf("Enter second number: ");
if (scanf("%lf", &num2) != 1) {
    printf("Invalid input. Please enter a numeric value.\n");
    return 1;
}

switch (choice) {
    case 1:
        operation = '+';
        result = num1 + num2;
        break;
    case 2:
        operation = '-';
        result = num1 - num2;
        break;
    case 3:
        operation = '*';
        result = num1 * num2;
        break;
    case 4:
        operation = '/';
        if (num2 == 0) {
            printf("\nError: Division by zero is not allowed.\n");
            return 1;
        }
        result = num1 / num2;
        break;
```

```
        default:

            printf("\nInvalid choice. Please select a number from 1 to 4.\n");

            return 1;

    }

    printf("Result: %.2lf %c %.2lf = %.2lf\n", num1, operation, num2, result);

    return 0;

}
```

Output:

Basic Calculator

Select an operation:

1. Addition(+)
2. Subtraction(-)
3. Multiplication(*)
4. Division(/)

Enter choice (1-4): 4

Enter first number: 12

Enter second number: 5

Result: 12.00 / 5.00 = 2.40